

BTEC Level 3 National Foundation. Extended Diploma in Mechanical Engineering.

Unit 25: Mechanical Behaviour of Metallic Materials.

Assignment 1: Investigation of Ferrous and Non-Ferrous Metallic Materials.

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Submission Date: 25th July, 2026.

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1. INTRODUCTION.

1.1 Aim.

The aim of this investigation is to analyse the microstructures of ferrous and non-ferrous metallic materials and determine how composition, processing history, grain boundaries and impurities influence mechanical properties.

Metals rarely fail or perform in service because of their chemical composition alone. Their behaviour is strongly influenced by features that exist on a microscopic scale, including grain size, phase distribution, crystal structure and the presence of impurities. These features control important engineering properties such as strength, hardness, ductility, toughness and resistance to fatigue.

By examining prepared micrographs and comparing them with recognised materials databases, it is possible to identify both the material and the manufacturing processes that have been applied to it. Understanding these relationships allows engineers to select appropriate materials for specific applications and predict how components will perform under service conditions.

This investigation examines a range of ferrous and non-ferrous materials in both processed and non-processed conditions. Particular attention is given to the effects of heat treatment, mechanical working, grain refinement and alloying additions on the resulting microstructures and mechanical properties.

1.2 Objectives.

The objectives of this investigation are to:

- Identify metallic materials through the examination of metallographic micrographs.
- Compare observed microstructures with accredited reference sources to confirm material identification.
- Determine the processing history of each specimen by analysing characteristic microstructural features.
- Examine the relationship between grain structure and mechanical behaviour.
- Investigate the influence of impurities and inclusions on material performance.
- Assess how manufacturing processes such as annealing, normalising, hot rolling, cold rolling and quenching modify microstructures.
- Evaluate how changes in microstructure affect strength, hardness, ductility, toughness and fatigue resistance.
- Relate the observed microstructures to real engineering applications and service requirements.

1.3 Engineering Relevance.

The study of metallic microstructures is a key aspect of materials engineering because the performance of a component is directly linked to its internal structure. Engineers must understand how materials respond to processing and service conditions in order to ensure safety, reliability and cost-effective design.

Aerospace Engineering.

Aircraft structures operate under demanding conditions involving cyclic loading, vibration and significant temperature variations. Materials used in aerospace applications require a combination of high strength, low weight and excellent fatigue resistance. The microstructure of these materials must be carefully controlled to prevent crack initiation and ensure long service life. Even small changes in grain size or phase distribution can significantly affect performance in critical aerospace components.

Marine Engineering.

Marine structures are exposed to corrosive saltwater environments, fluctuating loads and impact forces generated by waves and operating conditions. Materials selected for ships, offshore platforms and marine equipment require a balance of strength, toughness and corrosion resistance. Microstructural examination helps engineers assess whether a material possesses the necessary characteristics to withstand these demanding environments without premature failure.

Hydraulic Systems.

Hydraulic cylinders, pumps and valves rely on materials that can withstand repeated pressure loading while maintaining dimensional stability and wear resistance. Components operating under high pressures must resist fatigue damage and deformation throughout their service life. The relationship between microstructure and mechanical properties is therefore critical when selecting materials for hydraulic applications.

Automotive Manufacturing.

The automotive industry uses a wide range of metallic materials to satisfy conflicting requirements such as strength, safety, fuel efficiency and manufacturing cost. Low-carbon steels may be selected for body panels because of their formability, while harder steels are used in gears, shafts and suspension components. Understanding microstructure allows engineers to tailor material properties through heat treatment and processing techniques to meet specific performance requirements.

The ability to identify microstructures and understand their effect on material behaviour is therefore essential for engineers involved in the design, manufacture and maintenance of engineering components. Accurate interpretation of micrographs provides valuable information about material quality, processing history and expected performance in service.

2. METHODOLOGY.

2.1 Sample Preparation.

Before a metallic material can be examined under a microscope, its surface must be carefully prepared to reveal its true microstructure. Any scratches, contamination or surface damage can obscure important features such as grain boundaries, phases and inclusions. Metallographic preparation therefore follows a series of controlled stages designed to produce a smooth and representative surface for examination.

Metallographic Lab Procedure.

To get a clear look at the internal grain structures without any surface scratches masking the phases, the metal samples had to go through a standard multi-stage metallographic preparation process. First, the specimen was carefully sectioned from the parent stock using an abrasive cutter. Plenty of liquid coolant was kept running over the blade during the cut to stop localized frictional heat from accidentally tempering or altering the underlying microstructure.

Once cut, the sample went onto the grinding wheels, moving progressively through finer grades of abrasive paper. The grinding direction was rotated exactly 90 degrees at each step to ensure that deep scratches left by the previous rough paper were entirely flattened out. After creating a uniform matte surface, the sample was polished using fine diamond compounds on a cloth wheel until it reached a completely mirror-like finish. At this stage, the surface reflects light uniformly and looks featureless under the microscope. To actually reveal the grain boundaries and distinct phases, the polished face was chemically etched using a specialized reagent, which selectively dissolves more reactive high-energy zones like grain boundaries and phase interfaces to create visual contrast.

The observed microstructures were then compared with accredited materials databases and reference micrographs in order to determine the material type and processing history.

Flowchart:

Sample Selection



Sectioning



Grinding



Polishing



Etching



Optical Microscopy



Microstructural Analysis

Figure 1. Typical metallographic preparation procedure used to prepare metallic samples for optical microscopy.

2.2 Etching Procedures.

Following polishing, I used different etchants depending on whether the specimen was ferrous or non-ferrous.

Ferrous Samples.

The steel and cast iron specimens were etched using 2% Nital, a solution consisting of nitric acid dissolved in ethanol.

Nital is one of the most widely used etchants for iron-based alloys because it provides excellent contrast between different phases within the microstructure. Ferrite, pearlite, cementite and martensite react differently to the solution, making them distinguishable under optical microscopy.

When I applied it for a controlled period, Nital preferentially attacked grain boundaries and phase interfaces. This produced visible contrast which allows the arrangement of grains and microstructural constituents to be examined in detail.

Non-Ferrous Samples.

The brass specimens were etched using a ferric chloride solution.

Ferric chloride is particularly effective for copper-based alloys because it reveals grain boundaries, annealing twins and deformation structures that are not visible on a polished surface. The etchant attacks different regions of the alloy at different rates, producing the contrast necessary for metallographic examination.

In annealed brass, ferric chloride clearly reveals recrystallised grains and annealing twins. In cold-worked brass, the etchant highlights elongated grains and deformation patterns produced during rolling.

Why Etching Reveals Grain Boundaries and Phases.

Grain boundaries are regions where adjacent crystals meet and possess different atomic orientations. These regions contain higher levels of stored energy compared with the interiors of the grains, making them more chemically reactive.

When an etchant is applied, grain boundaries are attacked more rapidly than the surrounding material. As a result, the boundaries appear darker under reflected light microscopy and become visible to the observer.

Different phases within a metallic alloy also possess variations in composition and crystal structure. Because each phase reacts differently to the etching solution, they reflect light differently after etching. This creates the contrast required to distinguish ferrite from pearlite, martensite from pearlite, or recrystallised grains from cold-worked grains.

The effectiveness of metallographic examination therefore depends heavily on correct etching procedures, as improper etching can either hide important features or create misleading structures that do not accurately represent the material.

3. FUNDAMENTAL METALLURGY.

Understanding the behaviour of metallic materials requires knowledge of their internal structure. Although metals may appear uniform to the naked eye, they are composed of crystals, grains, phases and defects that influence how they respond to loading, heat treatment and manufacturing processes. The mechanical properties observed in engineering materials are therefore closely related to their microstructure and atomic arrangement.

3.1 Crystal Structures.

The arrangement of atoms within a metal is known as its crystal structure. This arrangement affects how easily atoms can move when a force is applied and therefore influences properties such as strength, ductility and toughness.

Two of the most common crystal structures found in engineering materials are Face-Centred Cubic (FCC) and Body-Centred Cubic (BCC).

Face-Centred Cubic (FCC).

Examples:

- Brass
- Copper

In FCC materials, atoms are positioned at each corner of the cube and at the centre of each face. This arrangement provides a large number of slip systems, which are pathways that allow atomic layers to move past one another during deformation.

Because FCC metals contain many active slip systems, they generally exhibit excellent ductility and formability. This makes them suitable for manufacturing processes involving bending, drawing and rolling.

Copper and brass are commonly used in applications requiring significant deformation during manufacture because they can absorb strain without fracturing.

Body-Centred Cubic (BCC).

Examples:

- Ferritic Steel

In BCC materials, atoms are positioned at each corner of the cube and one atom is located in the centre. Compared with FCC structures, BCC materials have fewer easily activated slip systems, particularly at lower temperatures.

As a result, BCC metals generally possess higher strength but lower ductility than FCC metals. They are also more sensitive to temperature changes and can become brittle when exposed to sufficiently low temperatures.

Ferritic steels commonly used in structural applications possess a BCC crystal structure and demonstrate this behaviour.

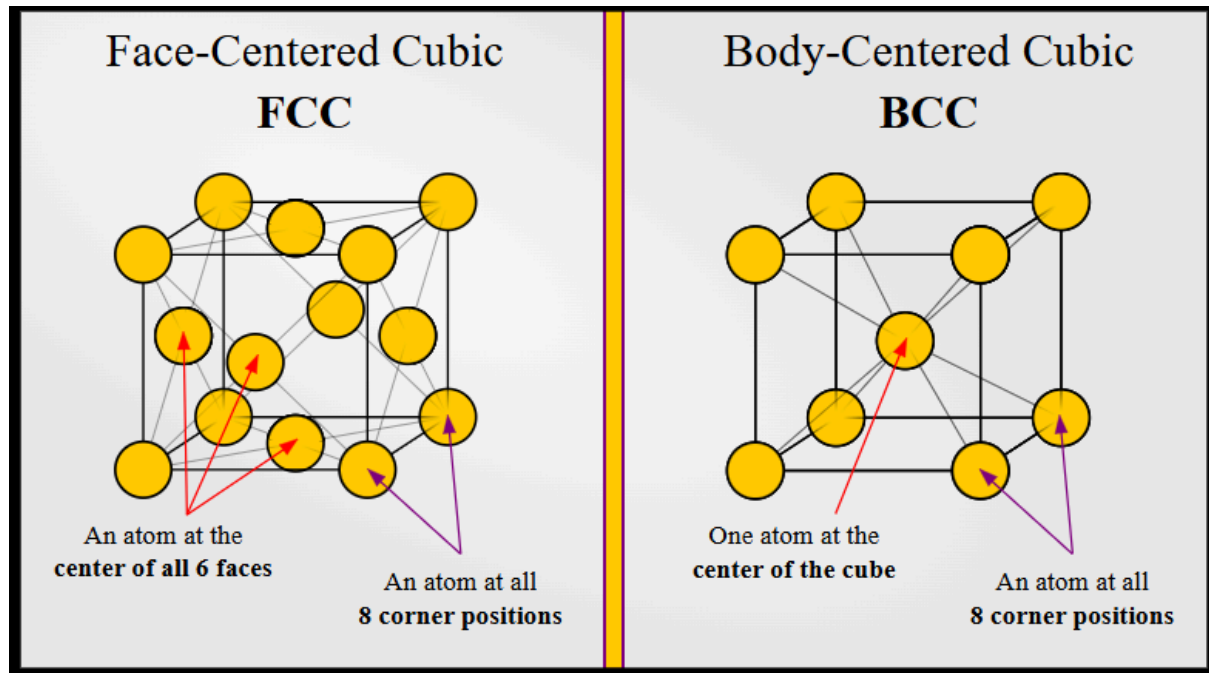


Figure 2. Comparison between FCC and BCC crystal structures showing atomic packing arrangements.

3.2 Dislocations and Plastic Deformation.

Although metals appear solid and continuous, they contain microscopic defects known as dislocations. These defects play a major role in determining how a material deforms under load.

Plastic deformation occurs when a material is permanently deformed after the applied stress exceeds its elastic limit. Rather than entire planes of atoms moving simultaneously, deformation occurs through the movement of dislocations.

Edge Dislocations.

An edge dislocation can be visualised as an extra half-plane of atoms inserted into the crystal structure.

When stress is applied, the dislocation moves through the crystal, allowing deformation to occur progressively rather than requiring all atoms to move at once. This significantly reduces the stress required for plastic deformation.

Screw Dislocations.

A screw dislocation is formed when atomic planes become displaced in a spiral arrangement around a central line.

Unlike edge dislocations, screw dislocations can move in multiple directions, making them important contributors to plastic deformation in many engineering alloys.

Slip Systems.

A slip system consists of a slip plane and a slip direction along which dislocations can move.

Materials with a greater number of active slip systems generally exhibit higher ductility because dislocations can move more easily throughout the crystal structure.

This explains why FCC metals such as copper and brass are usually more ductile than ferritic steels.

As dislocations move and multiply during deformation, they begin to interact with one another. These interactions make further movement increasingly difficult, resulting in strain hardening and an increase in material strength.

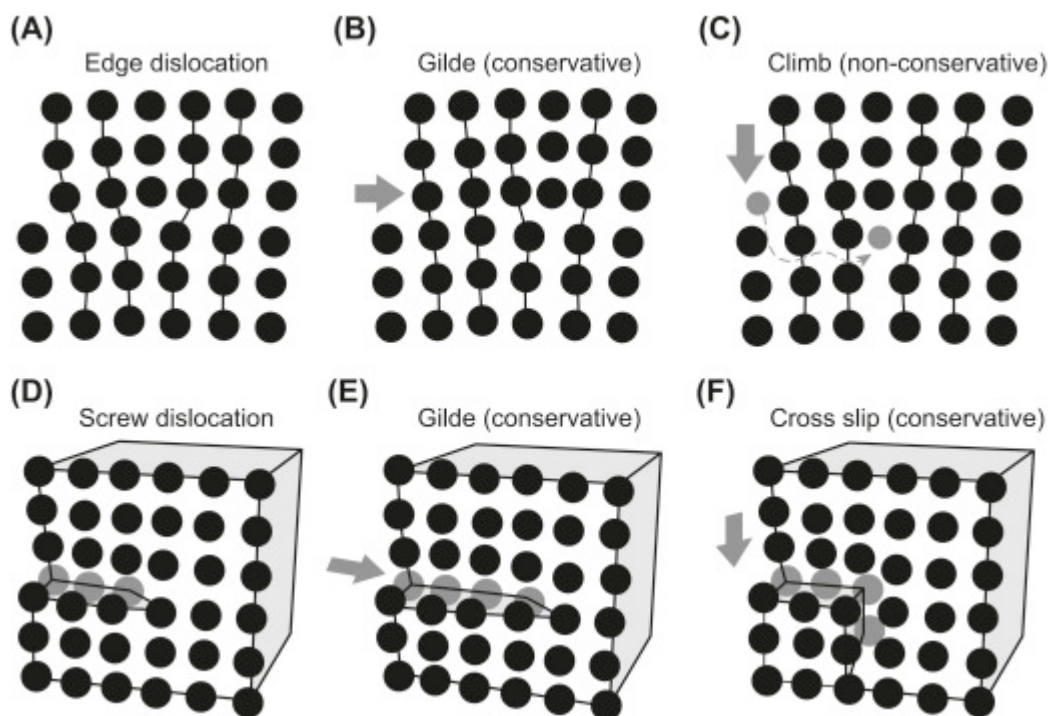


Figure 3. Schematic illustration of dislocation movement during plastic deformation.

3.3 Grain Boundaries and Hall-Petch Relationship.

Most engineering metals consist of many individual crystals known as grains. The regions where neighbouring grains meet are known as grain boundaries.

Grain boundaries act as obstacles to dislocation movement. When a dislocation reaches a grain boundary, it cannot easily continue into the adjacent grain because the crystal orientation has changed.

As a result, materials containing smaller grains possess a larger number of grain boundaries and therefore offer greater resistance to plastic deformation.

This relationship is described by the Hall-Petch equation:

$$\sigma_y = \sigma_0 + k_y / \sqrt{d}$$

Where:

σ_y = Yield Strength

σ_0 = Material Constant

k_y = Hall-Petch Constant

d = Average Grain Diameter

The equation shows that yield strength increases as grain size decreases.

Grain Refinement.

Grain refinement is achieved through controlled processing techniques such as normalising and controlled rolling.

Fine-grained structures generally provide:

- Higher strength
- Improved toughness
- Better fatigue resistance
- More uniform mechanical properties

Strength Increases.

The increased number of grain boundaries restricts dislocation movement, requiring greater stress to initiate plastic deformation.

Consequently, fine-grained materials are typically stronger than coarse-grained materials of the same composition.

Toughness.

Fine grain structures also improve toughness because cracks encounter more grain boundaries during propagation.

Each grain boundary changes the direction of crack growth and absorbs energy, reducing the likelihood of catastrophic fracture.

3.4 Solid Solution Strengthening.

Solid solution strengthening occurs when alloying elements are added to a metal and dissolve within its crystal lattice.

The added atoms distort the surrounding lattice structure, making dislocation movement more difficult and increasing strength.

Carbon in Steel.

Carbon atoms occupy interstitial spaces between iron atoms.

Although carbon atoms are relatively small, they create significant lattice distortion and greatly increase the strength and hardness of steel.

This is one reason why higher carbon steels generally possess greater strength than low-carbon steels.

Manganese in Shipbuilding Steel.

Manganese is commonly added to structural and shipbuilding steels to improve strength and toughness.

It contributes to solid solution strengthening while also helping to control the harmful effects of sulphur by forming manganese sulphide inclusions.

The manganese present in Specimen C contributes to its combination of strength, toughness and fatigue resistance.

Zinc in Brass.

Brass is produced by alloying copper with zinc.

The zinc atoms substitute for copper atoms within the lattice, creating distortion that restricts dislocation movement.

This strengthening effect allows brass to achieve greater strength than pure copper while retaining good corrosion resistance and formability.

3.5 Ductile-to-Brittle Transition Temperature.

Many BCC metals exhibit a phenomenon known as the ductile-to-brittle transition temperature (DBTT).

Above the transition temperature, the material behaves in a ductile manner and absorbs significant energy before fracture.

Below the transition temperature, dislocation movement becomes more difficult and the material may fracture suddenly with little or no plastic deformation.

This behaviour is particularly important in ferritic steels used in structural and marine applications.

For example, shipbuilding steels operating in cold environments must maintain adequate toughness to prevent brittle fracture. Historical failures of large steel structures have demonstrated the importance of selecting materials with suitable low-temperature performance.

The ductile-to-brittle transition is much less significant in FCC metals such as copper and brass because their numerous slip systems remain active even at lower temperatures.

Understanding crystal structures, dislocations, grain boundaries and strengthening mechanisms provides the foundation for interpreting the microstructures examined later in this investigation and explains why processing history has such a significant influence on mechanical properties.

4. PHASE DIAGRAMS AND MICROSTRUCTURAL DEVELOPMENT.

4.1 Iron-Carbon Phase Diagram.

The Iron-Carbon phase diagram is one of the most important tools used in metallurgy because it describes how the microstructure of steel changes as a function of temperature and carbon content. By studying the phase diagram, I can predict which phases will form during heating and cooling and how these phases will influence the mechanical properties of the material.

Steel is primarily composed of iron and carbon, with the carbon content determining the type and proportion of phases that develop during cooling. The main phases encountered in this investigation are ferrite, pearlite, cementite and austenite.

Ferrite (α -Iron).

Ferrite is a relatively soft and ductile phase with a Body-Centred Cubic (BCC) crystal structure. It contains very little dissolved carbon and therefore exhibits low hardness and high formability.

Low-carbon steels, such as Specimen A, contain large amounts of ferrite, which explains their excellent ductility and ease of forming.

Austenite (γ -Iron).

Austenite exists at elevated temperatures and possesses a Face-Centred Cubic (FCC) crystal structure. It can dissolve significantly more carbon than ferrite and plays an important role during heat treatment processes.

Although austenite is not normally present at room temperature in plain carbon steels, it acts as the parent phase from which pearlite and martensite are formed during cooling.

Cementite (Fe_3C).

Cementite is a hard and brittle iron carbide compound containing approximately 6.67% carbon. While cementite increases hardness and wear resistance, excessive amounts reduce toughness and ductility.

Pearlite.

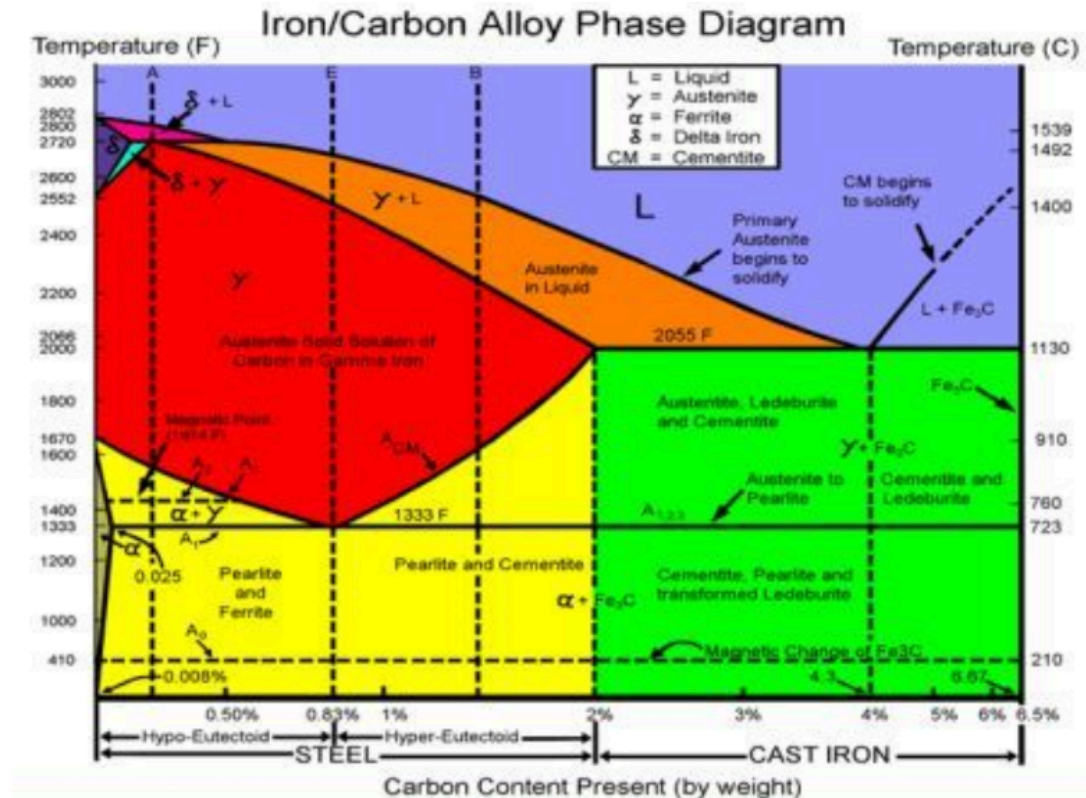
Pearlite is a lamellar mixture of ferrite and cementite formed when austenite transforms during slow cooling through the eutectoid temperature. The alternating layers of ferrite and cementite provide a balance of strength and ductility.

As carbon content increases, the proportion of pearlite within the microstructure also increases, resulting in higher strength and hardness.

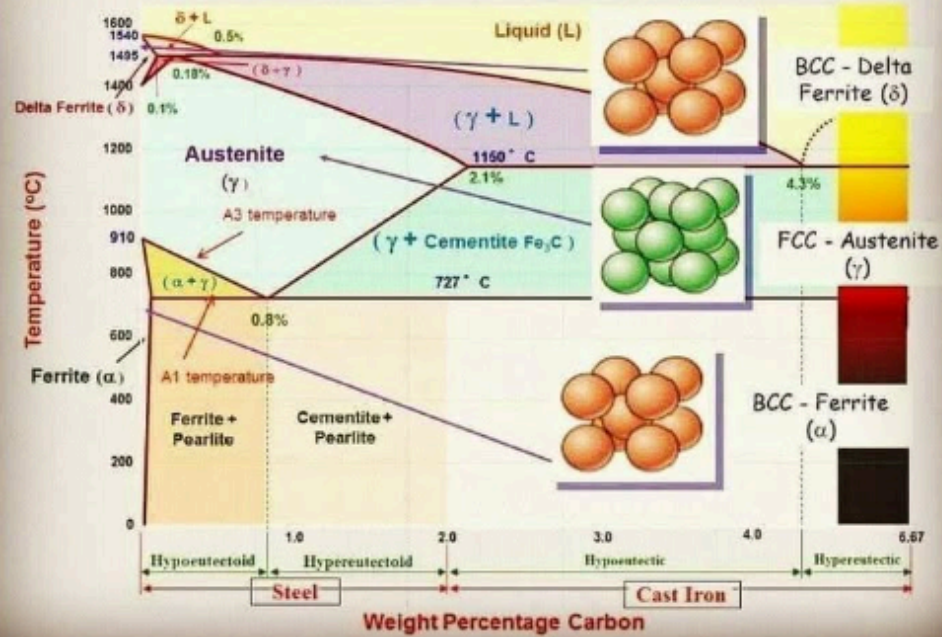
Eutectoid Composition.

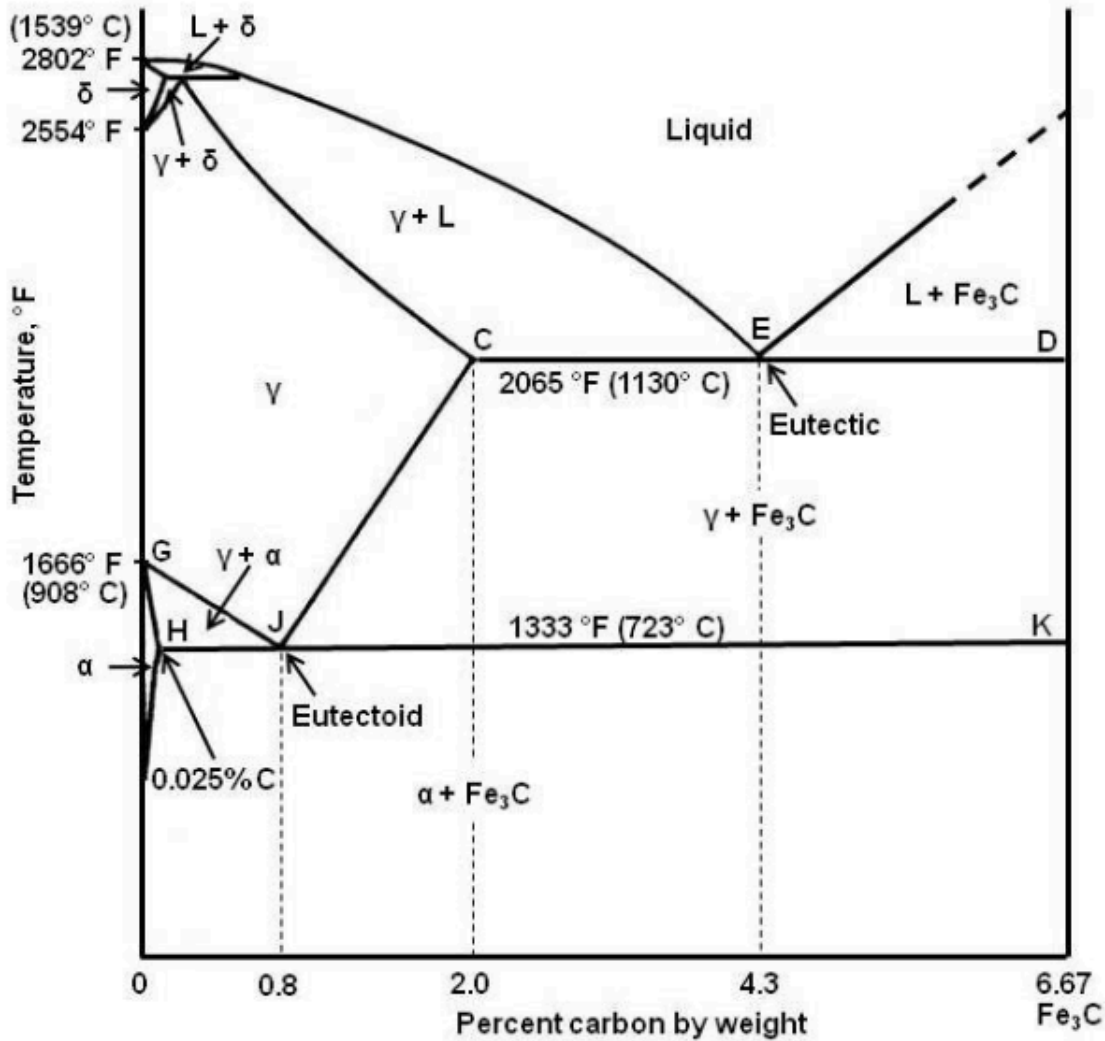
The eutectoid composition occurs at approximately 0.77% carbon and 727°C. At this point, austenite transforms completely into pearlite during cooling.

Steels containing less than 0.77% carbon are referred to as hypoeutectoid steels and contain both ferrite and pearlite. Steels containing more than 0.77% carbon are known as hypereutectoid steels and contain pearlite together with excess cementite.



Iron Carbon Binary Phase Diagram





The iron-iron carbide equilibrium diagram labeled in general terms

- Specimen A = 0.10% Carbon
- Specimen C = 0.16% Carbon
- Specimen B = 0.20% Carbon
- Specimen D = 0.90% Carbon

Figure 4. Iron-Carbon phase diagram showing carbon compositions of all steel specimens relative to the eutectoid composition.

Relationship Between Carbon Content and Microstructure

The specimens I examined in this investigation occupy different positions on the phase diagram and therefore develop different microstructures.

Specimen A contains approximately 0.10% carbon and is expected to consist predominantly of ferrite with a small amount of pearlite. This results in low strength but excellent ductility.

Specimen C contains approximately 0.16% carbon and exhibits a ferrite-pearlite structure. The addition of manganese provides additional strengthening while maintaining good toughness.

Specimen B contains approximately 0.20% carbon and develops a larger proportion of pearlite than Specimen A, resulting in increased hardness and strength.

Specimen D contains approximately 0.90% carbon, placing it above the eutectoid composition. Depending on the cooling rate, this steel may form pearlite during slow cooling or martensite during rapid quenching. Consequently, it exhibits the greatest variation in mechanical properties among the specimens examined.

4.2 Lever Rule Calculations.

I used the Lever Rule to estimate the proportion of phases present within an alloy under equilibrium cooling conditions. For hypoeutectoid steels, the Lever Rule allows the relative amounts of ferrite and pearlite to be calculated from the Iron-Carbon phase diagram.

Specimen B contains 0.20% carbon and therefore lies within the hypoeutectoid region.

Given:

Carbon content of steel = 0.20%

Carbon content of ferrite = 0.022%

Eutectoid composition = 0.77%

Calculation of Pearlite Fraction.

Fraction Pearlite

$$= (0.20 - 0.022) / (0.77 - 0.022)$$

$$= 0.178 / 0.748$$

$$= 0.238$$

Pearlite $\approx$ 23.8%

Calculation of Ferrite Fraction.

Fraction Ferrite

$$= (0.77 - 0.20) / (0.77 - 0.022)$$

$$= 0.57 / 0.748$$

$$= 0.762$$

Ferrite $\approx$ 76.2%

Predicted Microstructure.

The Lever Rule predicts that Specimen B should contain approximately:

- 76% Ferrite
- 24% Pearlite

This prediction agreed with the observed micrograph, where ferrite forms the majority phase and pearlite appears as darker colonies distributed throughout the structure.

The relatively high ferrite content explains why Specimen B retains reasonable ductility, while the increased pearlite content compared with Specimen A contributes to higher strength and hardness.

These calculations demonstrate how the Iron-Carbon phase diagram can be used not only to predict phase formation but also to estimate the relative proportions of microstructural constituents and their resulting influence on mechanical behaviour.

5. INVESTIGATION OF FERROUS MATERIALS.

5.1 Specimen A - 0.1% Carbon Steel.

Material Identification.

I identified Specimen A as an annealed low-carbon steel containing approximately 0.1% carbon. The microstructure consisted predominantly of ferrite with a small proportion of pearlite, which is consistent with the position of the material within the hypoeutectoid region of the Iron-Carbon phase diagram.

Processing History.

The large equiaxed ferrite grains indicate that the material had undergone a full annealing heat treatment. During annealing, the steel is heated above its critical temperature and then cooled slowly within the furnace. This slow cooling process allows new strain-free grains to form and grow, producing the coarse grain structure observed in the micrograph.

Microstructural Features.

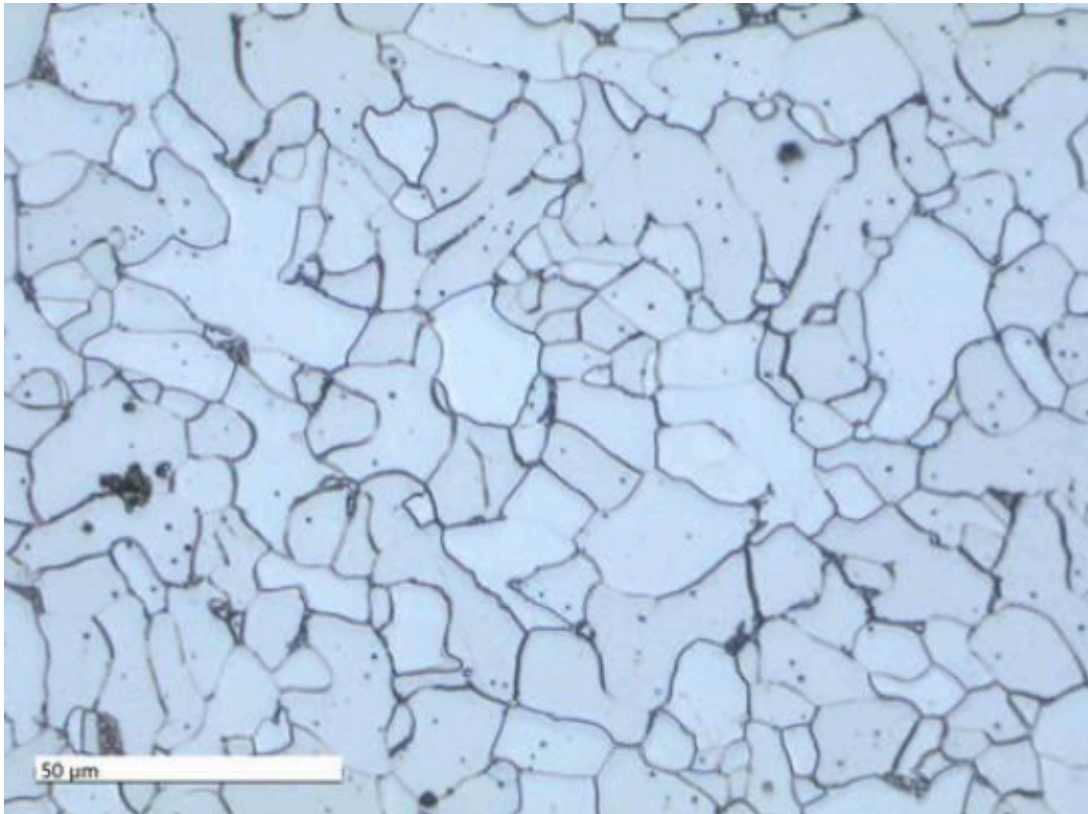


Figure 5. Optical micrograph of annealed 0.1% carbon steel showing equiaxed ferrite grains and grain boundaries.

The bright regions visible within the micrograph represent ferrite grains. The dark lines separating neighbouring grains are grain boundaries. The absence of significant pearlite indicates a very low carbon content.

Observation of the micrograph shows that the majority of the field of view consists of large, light-coloured ferrite grains separated by distinct grain boundaries. Very little pearlite is visible, which is consistent with the low carbon content of the steel. The relatively large grain size suggests slow cooling following heat treatment, supporting the conclusion that the specimen was fully annealed.

Mechanical Properties.

The predominantly ferritic structure results in:

- High ductility
- Low hardness
- Excellent formability
- Good machinability
- Low tensile strength compared with higher-carbon steels

Evaluation.

The final evaluation on this was that large ferrite grains provide relatively little resistance to dislocation movement. Consequently, the material deformed easily under load, resulting in excellent formability but reduced strength. This microstructure is therefore suitable for applications where shaping and forming are more important than maximum strength.

Typical applications include automotive body panels, structural sheet steel and fabricated components.

5.2 Specimen B - 0.2% Carbon Steel.

Material Identification.

I identified Specimen B as a ferrite-pearlite steel containing approximately 0.2% carbon. Compared with Specimen A, a larger proportion of pearlite was present due to the increased carbon content.

Processing History.

The relatively refined grain structure suggests that the material has been normalised. Normalising involves heating the steel above the critical temperature followed by cooling in still air. This produces a finer grain structure than annealing and improves strength and toughness.

Microstructural Features.



Figure 6. Optical micrograph of ferrite-pearlite steel showing pearlite colonies within a ferrite matrix.

The lighter regions correspond to ferrite while the darker colonies represent pearlite. Pearlite consists of alternating layers of ferrite and cementite that form during cooling from austenite.

Compared with Specimen A, a noticeably greater number of dark pearlite colonies can be observed throughout the microstructure. These colonies are distributed between the lighter ferrite regions and indicate an increase in carbon content. The finer appearance of the structure also supports the conclusion that the steel was normalised rather than fully annealed.

Mechanical Properties.

- Increased strength.
- Increased hardness.
- Improved wear resistance.
- Reduced ductility compared with Specimen A.

Evaluation.

My final evaluation on this was, the increased proportion of pearlite provides greater resistance to dislocation movement because the cementite lamellae act as barriers to deformation. Consequently, Specimen B possesses greater strength and hardness than Specimen A while retaining useful ductility.

The calculated ferrite and pearlite fractions obtained using the Lever Rule supports the observed microstructure and demonstrates the relationship between carbon content and mechanical behaviour.

This material is commonly used in shafts, brackets and general engineering components.

5.3 Specimen C - Shipbuilding Steel.

Specimen C: Structural Shipbuilding Steel.

Analyzing Specimen C reveals a low-carbon manganese steel specifically tailored for marine environments. The standout feature in this micrograph is the prominent pearlite banding, where dark colonies of pearlite are visibly stretched out and aligned in parallel tracks across the lighter ferrite matrix. This heavy directional alignment is a clear indicator of hot rolling during production. The steel is squashed through heavy rollers above its recrystallization temperature, elongating the underlying microstructure along the direction of travel.

Adding roughly 1.5% manganese to this mix is a deliberate engineering choice. It drives solid solution strengthening within the lattice and reacts with dangerous impurity elements like sulfur, locking them away as harmless manganese sulfide inclusions instead of brittle iron sulfides.

Microstructural Features.

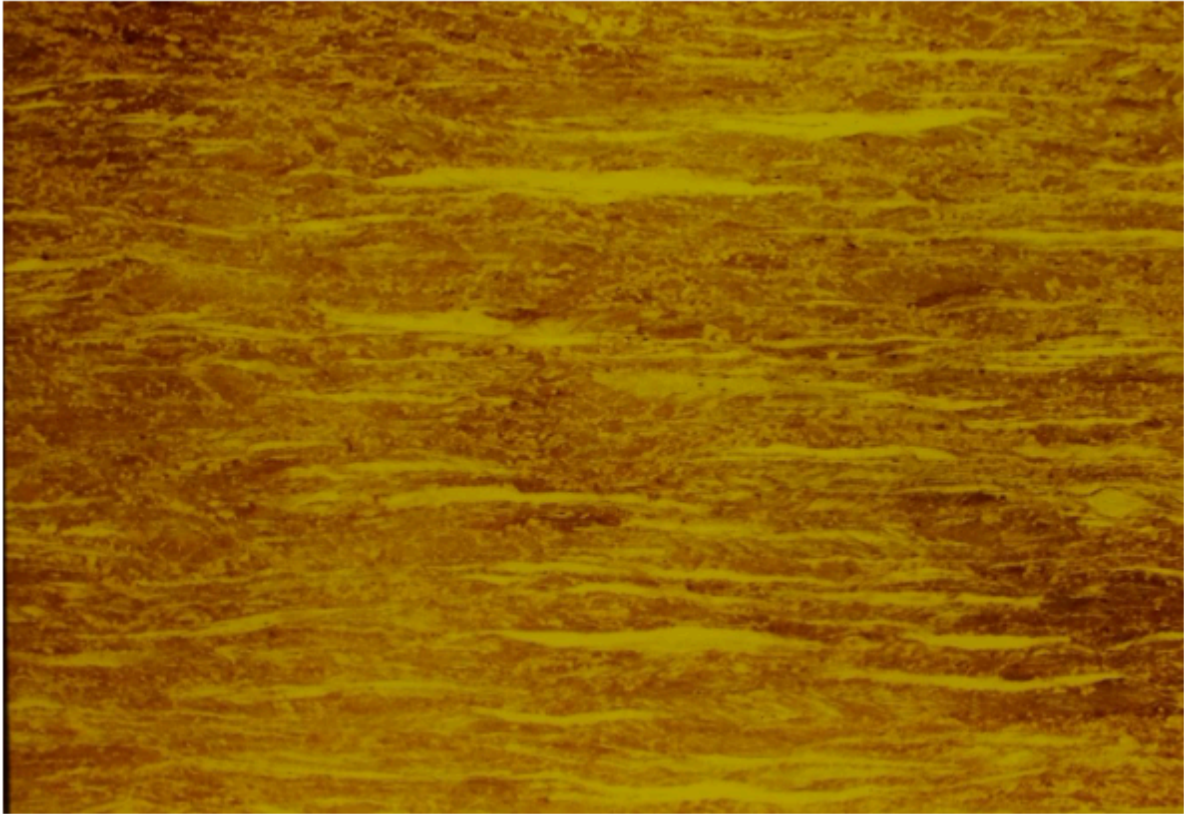


Figure 7. Optical micrograph of shipbuilding steel showing pearlite banding aligned with rolling direction.

This microstructural blending gives Specimen C an exceptionally reliable balance of yield strength, impact toughness, and cyclic fatigue resistance. It handles structural loading and wave impacts without cracking under pressure, which is exactly why it is chosen for structural ship hulls and offshore marine platforms.

Among the steels examined, Specimen C offers one of the most balanced combinations of mechanical properties and is therefore highly suitable for shipbuilding and structural engineering applications.

5.4 Specimen D1 - Furnace-Cooled 0.9% Carbon Steel.

Material Identification.

5.4 Specimen D1: Furnace-Cooled Eutectoid Steel (0.9% C).

Specimen D1 represents a high-carbon steel sitting just above the eutectoid threshold at roughly 0.9% carbon. Because this sample underwent an incredibly slow cooling cycle inside a hot furnace, the transforming austenite had ample time to undergo a complete, diffusion-controlled phase transformation near equilibrium. Under the microscope, this produces a clean, text-book lamellar microstructure made almost entirely of coarse pearlite colonies. You can clearly pick out the wide, alternating bands of soft ferrite and dark, brittle cementite lines.

Microstructural Features.



Figure 8. Furnace cooled high-carbon steel displaying coarse pearlite colonies.

Because cementite Fe_3C contains over 6.6% carbon, having this much of it tightly packed throughout the grains acts as a massive barrier to dislocation motion. When stress is applied, the hard carbide plates physically jam the slip planes, boosting the bulk material's hardness and wear resistance significantly. The major downside to this pearlite distribution is a sharp drop in ductility and impact energy absorption. While it's perfect for tools or rails that need to resist surface wear, it is too brittle to tolerate sudden impact or shock loading without fracturing cleanly along the lamellae.

5.5 Specimen D2 - Water-Quenched 0.9% Carbon Steel.

Specimen D2: Water-Quenched Martensitic Steel (0.9% C).

Specimen D2 provides a fascinating comparison because it shares an identical 0.9% carbon chemistry with D1, but a rapid water quench completely alters its internal structure. By plunging the white-hot steel straight into cold water, the cooling rate was accelerated so fast that normal carbon diffusion was totally suppressed. The carbon atoms never had the chance to step out and form pearlite plates. Instead, they got trapped mid-transformation inside the changing iron crystal lattice, forcing a diffusionless shift into an intensely strained, body-centered tetragonal structure known as martensite.

Visually, the coarse pearlite is entirely replaced by a highly dense, needle-like acicular network. This heavily distorted lattice structure completely locks down the slip systems, making plastic deformation nearly impossible. As a direct result, Specimen D2 hits maximum tensile strength and exceptional scratch hardness. The trade-off is extreme brittleness; the internal residual stress is so massive that the steel is highly prone to sudden, catastrophic cleavage failure under tensile stress. In a real workshop environment, this structure would absolutely require a subsequent tempering cycle to restore a safe level of engineering toughness before being put into service.

Microstructural Features.

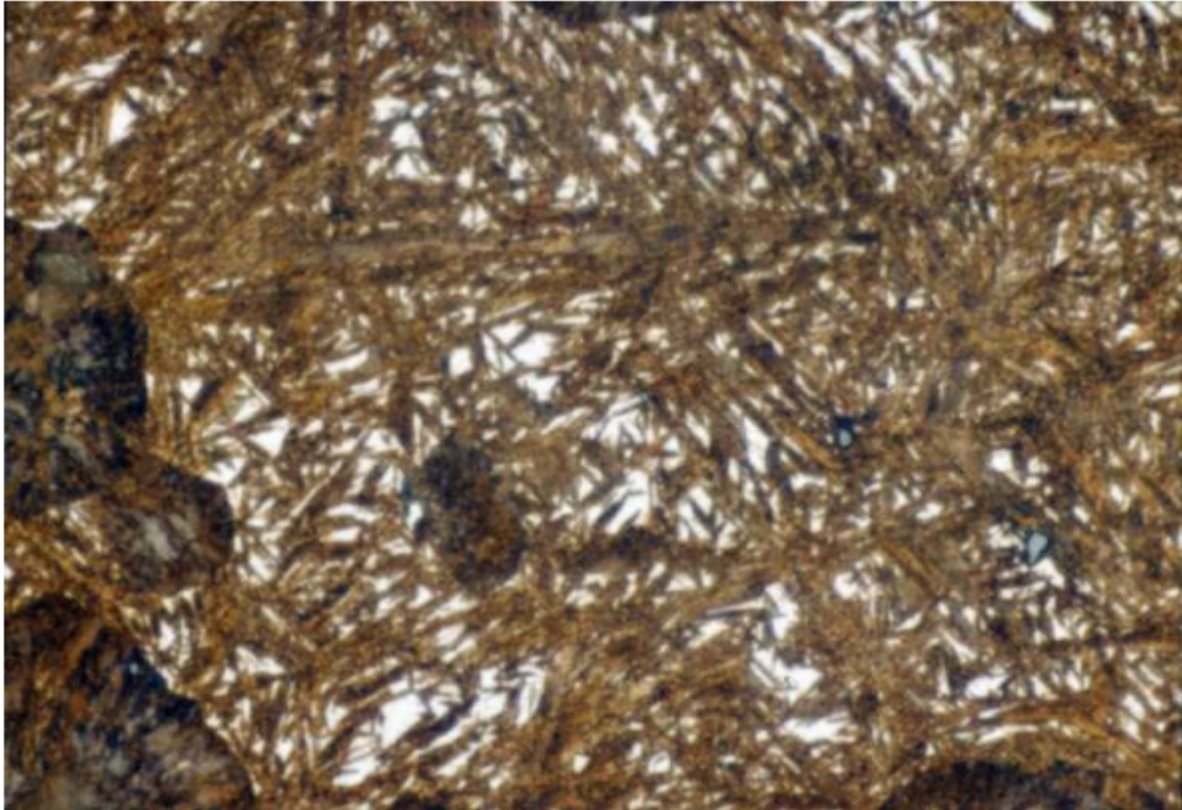


Figure 9. Water-quenched high-carbon steel showing acicular martensitic microstructure.

Comparison of Specimens D1 and D2 demonstrates how two samples with identical chemical compositions can possess dramatically different properties simply because of differences in cooling rate.

5.6 Specimen E - Spheroidal Cast Iron.

Material Identification.

Looking at the microstructure of Specimen E, this is a clear example of spheroidal graphite cast iron, which is commonly referred to in workshops as ductile iron. The standout feature here is the shape of the carbon. Instead of forming long strands, the graphite has balled up into distinct, rounded nodules scattered evenly across a bright, light-coloured ferritic matrix. This nodular shape is achieved during casting by adding a tiny amount of magnesium or cerium to the molten metal just before it is poured.

From an engineering perspective, this rounded shape changes everything compared to standard grey iron. Because the nodules are smooth and circular, they don't create sharp internal stress concentrations when the metal is pulled or loaded. The microstructural blending gives Specimen E excellent impact toughness, surprisingly good ductility, and high damping capacity. It means the material can flex and absorb sudden shock loads without snapping cleanly, making it the standard choice for high-pressure water mains, automotive crankshafts, and heavy-duty machinery brackets.

Microstructural Features.

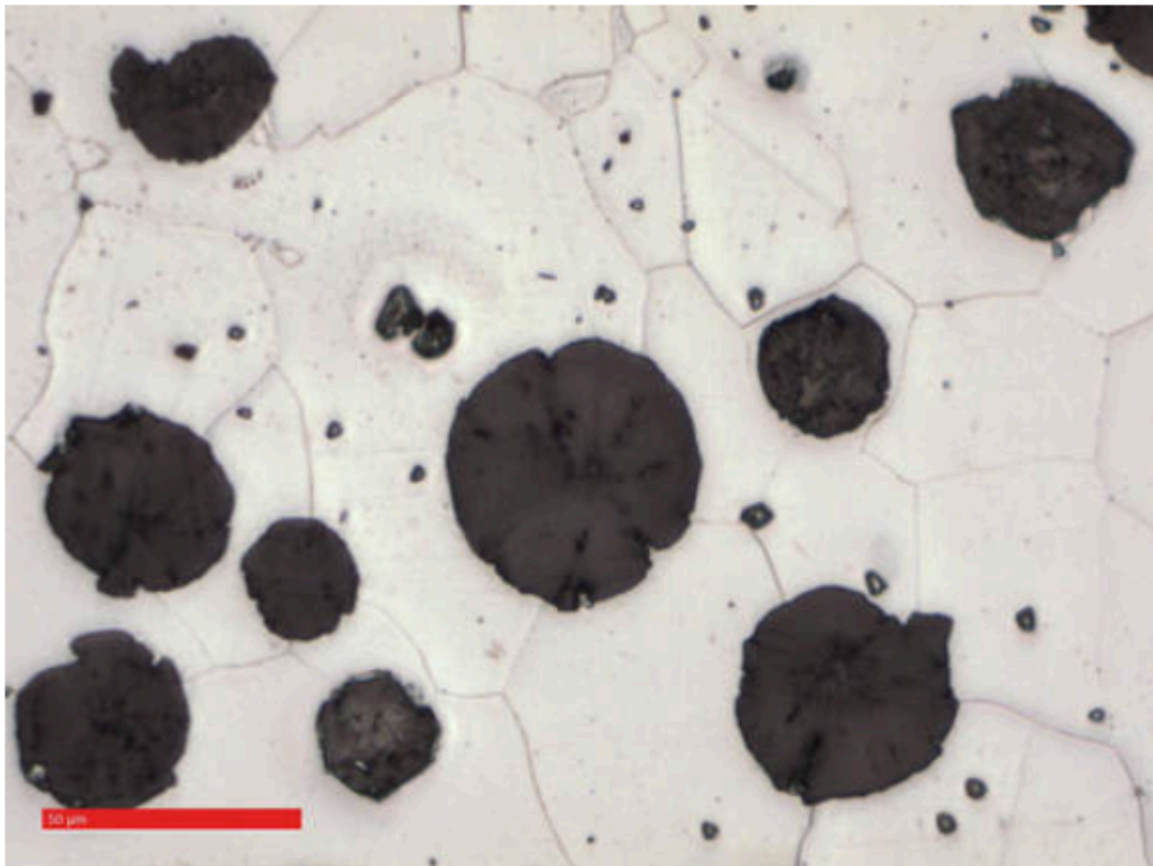


Figure 10. Ductile cast iron containing spheroidal graphite nodules dispersed within a ferritic matrix.

The graphite particles are predominantly spherical in shape and are distributed relatively uniformly throughout the ferritic matrix. The rounded morphology reduces stress concentration effects and explains the improved toughness associated with ductile cast iron.

5.7 Specimen H - Grey Cast Iron.

Material Identification.

Specimen H is a traditional grey cast iron, and it provides the perfect microstructural contrast to Specimen E. Instead of rounded nodules, the graphite here has formed a dense network of sharp, squiggly flakes embedded within a darker pearlitic matrix. Because this alloy is cooled relatively slowly without any special ladle treatments, the carbon naturally precipitates out into these long, thin interconnected ribbon shapes.

While this structure makes the iron incredibly easy to machine and gives it phenomenal compressive strength and vibration damping properties, it severely compromises its tensile performance. The sharp tips of these graphite flakes act like millions of tiny internal knife blades. The moment tensile or impact stress is applied, stress concentrates right at the tips of the flakes, causing microcracks to split open and propagate effortlessly through the material. This explains why grey iron behaves in a completely brittle manner and has very low toughness. However, because it absorbs vibrations so well and resists wear, it is still the absolute ideal material for engine blocks, brake discs, and heavy machine tool beds where components are compressed rather than pulled.

Microstructural Features.

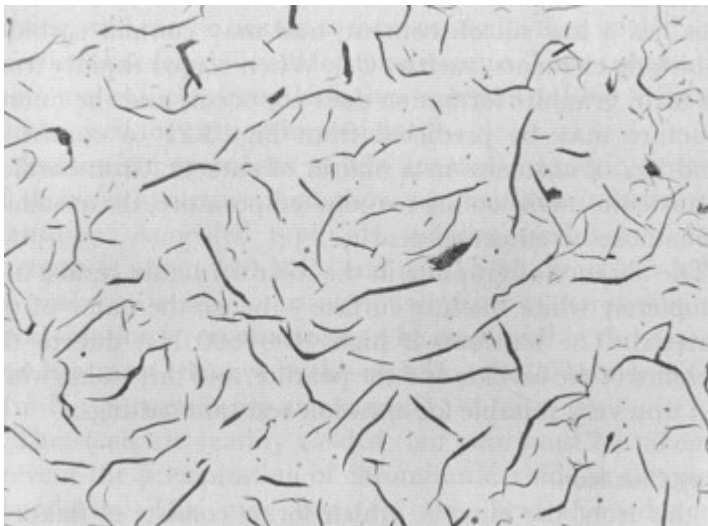


Figure 11. Grey cast iron contains graphite flakes which act as stress concentrators.

6. INVESTIGATION OF NON-FERROUS MATERIALS.

6.1 Specimen F - Annealed Brass.

Material Identification.

I identified Specimen F as annealed alpha brass. Alpha brass is a copper-zinc alloy containing up to approximately 35% zinc and possesses a single-phase Face-Centred Cubic (FCC) crystal structure. The FCC structure provides a large number of active slip systems, giving the material excellent ductility and formability.

Processing History.

The presence of equiaxed grains and numerous annealing twins indicates that the material has undergone an annealing heat treatment following previous deformation.

During annealing, the brass is heated to a temperature that promotes recrystallisation. New strain-free grains form and replace the deformed grains produced during mechanical working. As cooling occurs, a stable recrystallised structure develops, restoring ductility and reducing internal stresses.

Microstructural Features.



Figure 12. Annealed alpha brass showing recrystallised grains and annealing twins.

The micrograph reveals a uniform grain structure composed of equiaxed grains separated by clearly defined grain boundaries. Numerous straight lines crossing individual grains are visible and are identified as annealing twins.

Annealing twins form during recrystallisation and are a characteristic feature of many FCC materials including brass and copper. Their presence provides strong evidence that the material has undergone a heat treatment process rather than remaining in a cold-worked condition.

Several annealing twins can be observed crossing individual grains. These straight parallel features are characteristic of recrystallised FCC materials and provide strong evidence that the brass has undergone an annealing heat treatment.

Mechanical Properties.

The annealed microstructure results in:

- High ductility.
- Good toughness.
- Excellent formability.
- Reduced hardness.
- Reduced strength compared with cold-worked brass

Evaluation.

My final evaluation on this was the recrystallised grain structure allows dislocations to move relatively freely through the material. As a result, the brass can undergo significant plastic deformation before fracture.

The presence of annealing twins confirms that strain hardening effects have been removed through heat treatment. While this reduces strength and hardness, it significantly improves formability and makes the material suitable for manufacturing operations such as deep drawing, bending and forming.

Annealed brass is commonly used in plumbing fittings, decorative components, electrical connectors and heat exchanger applications where corrosion resistance and ease of fabrication are important.

6.2 Specimen G - Cold Rolled Brass.

Specimen G: Cold Rolled Brass.

Looking at Specimen G, we are dealing with an alpha-brass alloy that has been heavily cold-worked via rolling. Unlike the annealed specimen, this sample was rolled well below its recrystallization temperature, meaning the internal crystal grains were physically smashed, elongated, and flattened along the direction of processing. The micrograph shows highly distorted, squashed grains with no signs of the clean equiaxed structures or twin lines seen in heat-treated brass.

This mechanical cold work forces a rapid buildup of dislocations inside the face-centered cubic structure. As these dislocations multiply and slide into each other, they get tangled up and act as permanent internal roadblocks against further plastic slip. This phenomenon is strain hardening in action, and it drives up the yield strength and surface hardness of the brass significantly. The major engineering penalty here is a severe loss of ductility. Because the lattice is already so packed with flaws and internal stress, the material cannot stretch or deform much further without microcracks splitting open. This makes it a great choice for high-strength fasteners, stiff springs, or cartridge cases, but it cannot be used for any deep-drawing fabrication until it is annealed.

Microstructural Features.

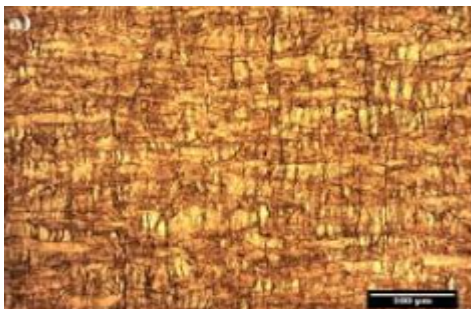


Figure 13. Cold rolled brass displaying elongated grains resulting from plastic deformation.

The micrograph shows elongated grains aligned in a preferred direction, indicating the influence of rolling. Unlike the equiaxed grains observed in Specimen F, the grains have been stretched and distorted by mechanical deformation.

The directional nature of the structure provides clear evidence of the manufacturing process applied to the material.

Comparison of Annealed and Cold Rolled Brass.

Property.	Annealed Brass (Specimen F).	Cold Rolled Brass (Specimen G).
Grain Shape	Equiaxed	Elongated
Dislocation Density	Low	High
Strength	Lower	Higher
Hardness	Lower	Higher
Ductility	Higher	Lower
Residual Stress	Low	High
Manufacturing Process	Annealed	Cold Rolled

The comparison demonstrates that mechanical processing can significantly alter material properties without changing chemical composition. The observed differences arise entirely from changes in microstructure and dislocation density produced during deformation and heat treatment.

7. PROCESSING HISTORY AND MECHANICAL PROPERTIES.

The mechanical properties of metallic materials are strongly influenced by their processing history. Although chemical composition plays an important role, manufacturing and heat treatment processes can significantly alter microstructure and therefore change the way a material responds to mechanical loading. The specimens examined in this investigation demonstrate how different processing routes affect strength, hardness, ductility and toughness.

Annealing.

Annealing is a heat treatment process in which a material is heated to an elevated temperature and then cooled slowly, usually within a furnace. This process promotes recrystallisation and grain growth, producing a microstructure containing relatively large equiaxed grains.

The annealed low-carbon steel (Specimen A) and annealed brass (Specimen F) both displayed microstructures characteristic of recrystallised materials. In each case, the grains appeared relatively uniform and strain-free, indicating that previous deformation had been removed.

The primary effect of annealing is an increase in ductility and toughness. Because the dislocation density is reduced, dislocations can move more easily through the crystal structure. As a result, the material becomes easier to form and shape. However, this improvement in ductility is accompanied by a reduction in strength and hardness.

Annealing is therefore commonly used when components require extensive forming or when residual stresses must be removed following manufacturing operations.

Hot Rolling.

Hot rolling is performed above the recrystallisation temperature of the material. During rolling, the material is plastically deformed while new grains continuously form and replace deformed structures.

Specimen C displayed clear evidence of hot rolling through the presence of pearlite banding aligned in the rolling direction. The resulting microstructure provides a useful balance between strength and toughness.

Compared with annealed materials, hot-rolled steels generally exhibit greater strength while maintaining acceptable ductility. Grain refinement produced during processing improves resistance to crack propagation and contributes to good fatigue performance.

These characteristics explain why hot-rolled steels are widely used in structural engineering, shipbuilding and large load-bearing components.

Cold Rolling.

Cold rolling is performed below the recrystallisation temperature and produces significant plastic deformation within the material. Unlike hot rolling, the deformed grain structure is retained after processing.

The cold-worked brass specimen exhibited elongated grains aligned in the rolling direction, indicating substantial deformation. During cold rolling, large numbers of dislocations are introduced into the crystal structure. These dislocations interact and obstruct one another, making further deformation increasingly difficult.

As a result, strength and hardness increase significantly through strain hardening. However, the material becomes less ductile because dislocation movement is restricted. Residual stresses may also develop within the material, which can increase susceptibility to distortion or cracking during service.

Cold rolling is therefore commonly used when increased strength is required without altering chemical composition.

Furnace Cooling.

Furnace cooling produces one of the slowest cooling rates used in heat treatment. The extended cooling period allows diffusion-controlled transformations to occur under near-equilibrium conditions.

Specimen D1, the furnace-cooled high-carbon steel, developed a coarse pearlitic microstructure consisting of alternating layers of ferrite and cementite. Because cooling occurs slowly, the phases have sufficient time to form well-defined lamellar structures.

The resulting material exhibits increased hardness and strength compared with low-carbon steels, largely due to the presence of cementite. However, toughness and ductility are reduced because the hard cementite phase restricts plastic deformation.

Furnace cooling therefore produces a relatively stable microstructure with a useful combination of wear resistance and mechanical strength.

Quenching.

Quenching involves rapid cooling from the austenitic region, typically using water, oil or another cooling medium. The cooling rate is sufficiently high to suppress normal diffusion-controlled transformations.

Specimen D2 demonstrated the effects of quenching through the formation of martensite. The needle-like martensitic structure develops because carbon atoms become trapped within the distorted crystal lattice during rapid cooling.

Among all specimens examined, the quenched steel exhibited the highest hardness and strength. The highly distorted crystal structure severely restricts dislocation movement, making plastic deformation extremely difficult.

However, the increase in hardness comes at the expense of ductility and toughness. Martensitic steels can be susceptible to brittle fracture if used without subsequent tempering.

Overall Comparison.

The specimens I examined throughout this investigation demonstrated that processing history has a major influence on mechanical behaviour. Annealing increases ductility and toughness by reducing dislocation density, while cold rolling increases strength through strain hardening. Hot rolling provides a balance of strength and toughness through grain refinement and controlled deformation. Furnace cooling promotes the formation of pearlite, producing moderate hardness and wear resistance, whereas quenching generates martensite and results in maximum hardness and strength.

The comparison clearly shows that materials with similar compositions can exhibit dramatically different mechanical properties depending on the manufacturing and heat treatment processes applied. Consequently, understanding processing history is essential when selecting materials for engineering applications and predicting their performance in service.

8. GRAIN BOUNDARIES, IMPURITIES AND FAILURE MECHANISMS.

8.1 Grain Boundary Strengthening.

Grain boundaries are the interfaces that separate individual crystals, or grains, within a metallic material. Although they occupy only a small proportion of the total structure, they have a significant influence on mechanical behaviour.

When a metal is subjected to stress, plastic deformation occurs through the movement of dislocations. As dislocations travel through a grain, they eventually encounter a grain boundary. Because neighbouring grains possess different crystal orientations, the dislocation cannot easily continue into the adjacent grain. The grain boundary therefore acts as a barrier to dislocation movement.

This strengthening effect is described by the Hall-Petch relationship:

$$\sigma_y = \sigma_0 + k_y / \sqrt{d}$$

Where:

σ_y = Yield Strength

σ_0 = Material Constant

k_y = Hall-Petch Constant

d = Average Grain Diameter

The equation demonstrates that reducing grain size increases yield strength. Fine-grained materials contain a greater number of grain boundaries and therefore provide increased resistance to plastic deformation.

The effect can be observed by comparing materials with coarse annealed structures to those that have undergone controlled rolling or normalising. Finer grain structures generally exhibit greater strength and improved toughness because grain boundaries also impede crack propagation.

As a result, grain refinement is widely used in engineering materials where a balance of strength, toughness and fatigue resistance is required.

8.2 Impurity Segregation.

No engineering material is completely pure. During manufacturing, small quantities of impurities and non-metallic inclusions become incorporated into the material. Although often present in low concentrations, these features can significantly influence mechanical performance.

Sulphur.

Sulphur is generally considered a harmful impurity in steel. When present in excessive quantities, it can combine with iron to form iron sulphide compounds.

Iron sulphide possesses a relatively low melting temperature and may accumulate along grain boundaries. During hot working operations, these regions can become weak and susceptible to cracking, a phenomenon historically known as hot shortness.

For this reason, sulphur levels are carefully controlled in structural steels.

Oxides.

Oxide inclusions form when oxygen reacts with metallic elements during steelmaking and casting processes.

These inclusions are generally hard and brittle compared with the surrounding metal. Under loading conditions, oxides may act as local stress concentrators and become sites for crack initiation.

Large oxide inclusions can reduce fatigue performance and decrease overall toughness.

Silicates.

Silicate inclusions originate from slag particles or reactions occurring during steel production.

Like oxide inclusions, silicates do not deform as easily as the surrounding metal matrix. Their presence can disrupt the uniform distribution of stress and contribute to premature crack formation under cyclic loading conditions.

The size, shape and distribution of silicate inclusions strongly influence their effect on mechanical performance.

Manganese Sulphides.

Manganese is commonly added to steels to reduce the harmful effects of sulphur.

Instead of forming iron sulphide, sulphur combines preferentially with manganese to produce manganese sulphide inclusions. These inclusions are generally less harmful and help improve machinability.

In structural steels such as shipbuilding steel, manganese provides both solid solution strengthening and improved control of sulphur-related defects.

The manganese content present in Specimen C contributes to its combination of strength, toughness and reliability.

Although individual inclusions were not clearly visible within all micrographs, their influence remains important when evaluating engineering performance. Even small concentrations of impurities can significantly affect fatigue life, fracture behaviour and long-term reliability.

8.3 Fatigue Failure.

Fatigue is one of the most common causes of failure in engineering components. Unlike overload failure, fatigue can occur at stress levels significantly below the yield strength of a material.

Repeated cyclic loading gradually produces microscopic damage that accumulates over time until fracture eventually occurs.

The fatigue process generally occurs in three stages:

Crack Initiation.

Fatigue cracks often begin at stress concentrators such as:

- Surface defects
- Scratches
- Inclusions
- Sharp corners
- Grain boundary discontinuities

These regions experience locally elevated stresses which promote crack formation.

Crack Propagation.

Once initiated, the crack grows incrementally during each loading cycle.

The crack may remain undetected for a considerable period while progressively reducing the effective cross-sectional area of the component.

Final Fracture.

When the remaining material can no longer support the applied load, rapid fracture occurs.

Although the final fracture appears sudden, the majority of the component's life is typically spent in the crack initiation and propagation stages.

Applied Cyclic Stress



Crack Initiation



Crack Propagation



Final Fracture

Figure 14. Stages of fatigue failure showing crack initiation, propagation and final fracture.

The fatigue performance of a material is strongly influenced by grain size, inclusions, surface condition and manufacturing quality. Fine-grained materials generally exhibit improved fatigue resistance because grain boundaries impede crack growth.

8.4 Stress Corrosion Cracking in Brass.

Stress corrosion cracking is a failure mechanism that occurs when a susceptible material is simultaneously exposed to tensile stress and a corrosive environment.

One of the most well-known examples is season cracking in brass.

Season cracking was first observed in brass cartridge cases stored in environments containing ammonia. Residual stresses generated during cold working combined with chemical attack from the surrounding environment, resulting in the formation of intergranular cracks.

These cracks often propagated along grain boundaries and could lead to sudden failure despite relatively low applied stresses.

Cold-worked brass is particularly susceptible because deformation introduces residual stresses and increases stored strain energy within the microstructure.

The risk can be significantly reduced through stress-relief annealing, which removes residual stresses and restores a more stable microstructure.

The comparison between annealed brass (Specimen F) and cold-worked brass (Specimen G) demonstrates why processing history is important when considering resistance to environmentally assisted cracking.

8.5 Grain Boundary Embrittlement.

Although grain boundaries often improve strength and toughness, they can also become sites of weakness under certain conditions.

Grain boundary embrittlement occurs when impurities segregate to grain boundaries and reduce the cohesion between neighbouring grains.

Elements such as sulphur, phosphorus and other trace impurities may preferentially accumulate along grain boundaries during processing or heat treatment. These segregated regions become mechanically weaker than the surrounding grains.

When stress is applied, cracks can propagate along the weakened boundaries rather than through the grains themselves. This produces intergranular fracture, which is generally characterised by low toughness and limited plastic deformation.

Materials affected by grain boundary embrittlement may fail unexpectedly even when the bulk material appears to possess adequate strength.

The phenomenon highlights the importance of controlling impurity levels during material production and demonstrates how microscopic features can significantly influence macroscopic mechanical behaviour.

Overall, the investigation of grain boundaries, impurities and failure mechanisms demonstrates that material performance is governed not only by chemical composition but also by microstructural features developed during processing. Understanding these relationships enables engineers to select suitable materials, design safer components and minimise the risk of premature failure in service.

9. ACCREDITED SOURCE VALIDATION.

The identification of each specimen was verified through comparison with recognised metallurgical reference sources, including the University of Cambridge DoITPoMS (Dissemination of IT for the Promotion of Materials Science) database and MatWeb Materials Property Data. These sources provide validated micrographs, material descriptions and mechanical property information that can be used to confirm metallographic observations.

Comparison of the observed microstructures with accredited reference material showed a strong correlation between the specimens examined and the published micrographs. The agreement between grain morphology, phase distribution, graphite shape and deformation features provided confidence in the identification of each material and its associated processing history.

Specimen A - Annealed Low Carbon Steel.

The micrograph of Specimen A displayed large equiaxed ferrite grains separated by clearly defined grain boundaries. Only a small amount of pearlite was visible, indicating a very low carbon content.

Comparison with DoITPoMS ferritic steel micrographs showed the same characteristic grain structure associated with annealed low-carbon steels. MatWeb data for low-carbon structural steels also indicated high ductility, low hardness and excellent formability, which is consistent with the observed ferritic microstructure.

The agreement between the observed grain morphology and the reference sources confirms the identification of Specimen A as a fully annealed low-carbon steel.

Specimen B - Ferrite-Pearlite Steel.

Specimen B contained a mixture of light ferrite regions and darker pearlite colonies distributed throughout the microstructure.

Reference micrographs I obtained from DoITPoMS showed similar ferrite-pearlite distributions for hypoeutectoid steels containing approximately 0.2% carbon. The predicted ferrite and pearlite fractions calculated using the Lever Rule were also consistent with the observed structure.

MatWeb property data indicated increased strength and hardness relative to lower-carbon steels, which corresponds with the greater proportion of pearlite observed within the specimen.

The combined evidence confirms the identification of Specimen B as a normalised ferrite-pearlite steel.

Specimen C - Shipbuilding Steel.

The most distinctive feature of Specimen C was the presence of pearlite banding aligned in a preferred direction. This banding is characteristic of hot-rolled structural steels.

DoITPoMS reference images I added of rolled low-carbon manganese steels displayed similar banded structures resulting from mechanical processing. The addition of manganese and the observed ferrite-pearlite matrix correspond closely with material data published for shipbuilding steels.

The observed microstructure, processing indicators and expected mechanical properties confirm the identification of Specimen C as a hot-rolled manganese-containing shipbuilding steel.

Specimen D1 - Furnace-Cooled High Carbon Steel.

Specimen D1 displayed coarse pearlite colonies composed of alternating ferrite and cementite lamellae.

Comparison with pearlitic steel micrographs available through accredited metallurgical references showed very similar lamellar structures associated with slow cooling from the austenitic region. The relatively coarse spacing between the lamellae supports the conclusion that furnace cooling occurred at a slow rate.

The observed microstructure agrees closely with published examples of furnace-cooled high-carbon steel and therefore confirms the identification.

Specimen D2 - Water-Quenched High Carbon Steel.

The micrograph of Specimen D2 contained needle-like and acicular features typical of martensite.

DoITPoMS reference images I added of quenched steels displayed nearly identical martensitic structures. Published descriptions emphasised the characteristic needle morphology produced when diffusion-controlled transformations are suppressed during rapid cooling.

MatWeb data for quenched high-carbon steels also indicated very high hardness and strength combined with reduced ductility, matching the expected behaviour of the observed microstructure.

These observations confirm that Specimen D2 is a water-quenched martensitic steel.

Specimen E - Ductile Cast Iron.

Specimen E contained numerous dark graphite nodules dispersed throughout a ferritic matrix.

Reference images of ductile cast iron I added consistently show spheroidal graphite particles with similar size and distribution. Published mechanical property data also demonstrate improved toughness and ductility compared with grey cast iron due to the rounded graphite morphology.

The close agreement between the observed graphite nodules and accredited reference micrographs confirms the identification of Specimen E as ductile cast iron.

Specimen F - Annealed Alpha Brass.

Specimen F exhibited equiaxed grains together with numerous annealing twins, which are characteristic features of recrystallised FCC alloys.

Comparison with reference brass micrographs sourced from DoITPoMS revealed almost identical twin structures commonly observed after annealing. Published property data describing annealed alpha brass also corresponded with the observed characteristics of high ductility and excellent formability.

The presence of annealing twins and recrystallised grains therefore confirms the identification of Specimen F as annealed alpha brass.

Validation Summary.

The comparison I did of all specimens with accredited metallurgical databases demonstrated strong agreement between observed microstructures and published reference materials. Characteristic features such as ferrite grains, pearlite colonies, martensitic needles, graphite nodules, annealing twins and rolling banding were consistently matched with recognised examples.

The close correlation between observed structures, expected processing histories and published mechanical property data provides confidence that the material identifications made throughout this investigation are accurate. This validation process reinforces the importance of using accredited reference sources when conducting metallographic analysis and ensures that conclusions drawn from the investigation are technically reliable.

The validation process demonstrated that the observed microstructures were not identified solely from visual appearance. Instead, grain morphology, phase distribution and processing indicators were systematically compared with accredited reference sources before final material classifications were assigned.

10. COMPARATIVE EVALUATION.

The specimens I examined throughout this investigation demonstrated how composition, processing history and microstructure influence engineering performance. Although no single material is superior in every category, comparison of the observed microstructures allows conclusions to be drawn regarding strength, ductility, hardness, brittleness and suitability for specific engineering applications.

Strongest Material.

The strongest material I examined was **Specimen D2 (Water-Quenched 0.9% Carbon Steel)**.

The microstructure consisted predominantly of martensite, identified by its needle-like acicular appearance. Martensite forms when austenite is rapidly quenched, trapping carbon atoms within a highly distorted crystal lattice. This distortion creates significant resistance to dislocation movement, resulting in exceptionally high strength.

Compared with all other specimens, D2 exhibited the greatest barrier to plastic deformation and therefore possessed the highest tensile strength.

However, the increase in strength is accompanied by reduced toughness and ductility.

Most Ductile Material.

The most ductile material was **Specimen F (Annealed Alpha Brass)**.

The micrograph displayed large recrystallised equiaxed grains together with numerous annealing twins. The FCC crystal structure of brass provides multiple active slip systems, allowing extensive plastic deformation before fracture.

Annealing removed the effects of strain hardening and reduced dislocation density, enabling dislocations to move relatively freely through the structure.

As a result, Specimen F demonstrated the greatest capacity for plastic deformation and would be expected to exhibit superior formability compared with all other specimens examined.

Hardest Material.

The hardest material was **Specimen D2 (Water-Quenched 0.9% Carbon Steel)**.

The martensitic structure produced by rapid quenching severely restricts dislocation movement due to lattice distortion caused by trapped carbon atoms.

This structure provides exceptional resistance to indentation and wear, making it significantly harder than pearlitic, ferritic or brass microstructures.

The comparison between D1 and D2 clearly demonstrates how cooling rate alone can dramatically increase hardness without changing chemical composition.

Most Brittle Material.

The most brittle material was **Specimen D2 (Water-Quenched 0.9% Carbon Steel)**.

Although grey cast iron is often regarded as a brittle engineering material, the martensitic steel exhibits extremely limited plastic deformation before fracture because of its highly strained crystal structure.

The absence of significant ductility means that failure may occur suddenly under impact or shock loading conditions.

This highlights the trade-off that often exists between hardness, strength and toughness.

If Specimen H (Grey Cast Iron) is included within the investigation, it may also be considered highly brittle due to the presence of graphite flakes that act as stress concentrators and crack initiation sites.

Best Aerospace Material.

The most suitable aerospace material among the specimens I examined is **Specimen F (Annealed Alpha Brass)** for applications requiring corrosion resistance and formability, although it would not normally be used for primary aircraft structures.

From the materials investigated, none represent modern aerospace alloys. However, brass offers excellent corrosion resistance, good fatigue behaviour and ease of manufacture.

For highly loaded aerospace structures, engineers would typically select aluminium or titanium alloys rather than the materials examined in this investigation.

Based solely on the available specimens, Specimen F provides the most balanced combination of corrosion resistance, ductility and manufacturing flexibility.

Best Marine Material.

The most suitable marine material was **Specimen C (Shipbuilding Steel)**.

The ferrite-pearlite microstructure, combined with manganese strengthening, provides an excellent balance of strength, toughness and fatigue resistance.

The observed pearlite banding confirmed a hot-rolled processing history, while the manganese content contributes to improved mechanical performance and structural reliability.

Marine structures experience cyclic loading, impact forces and harsh environmental conditions. Specimen C possesses the toughness required to resist crack propagation while maintaining adequate strength for large structural applications.

These characteristics explain why similar steels are widely used in ship hulls, offshore structures and marine engineering systems.

Best Overall Engineering Material.

The best overall engineering material was **Specimen C (Shipbuilding Steel)**.

While Specimen D2 possessed the highest strength and hardness, its low ductility and toughness limit its use in many engineering applications. Specimen F exhibited excellent ductility but lacked the strength required for heavily loaded structures.

Specimen C provided the most balanced combination of:

- Strength
- Toughness
- Fatigue resistance
- Manufacturability
- Structural reliability

Its ferrite-pearlite microstructure allows it to withstand significant loading while maintaining resistance to crack initiation and propagation. The addition of manganese further improves mechanical performance without excessively reducing ductility.

For this reason, Specimen C offers the most versatile combination of properties and would be the preferred choice for a wide range of engineering applications.

While Specimen D2 exhibited superior hardness and strength, its limited toughness reduces its suitability for many structural applications. Conversely, Specimen C provides a more balanced combination of properties, making it a safer and more versatile engineering material in real-world service conditions.

Overall Conclusion.

My investigation clearly demonstrated that microstructure has a major influence on engineering performance. The strongest and hardest material was the martensitic steel produced by quenching, while the most ductile material was annealed brass with its recrystallised FCC structure. Shipbuilding steel provided the most balanced combination of properties and was therefore judged to be the most effective overall engineering material.

The comparison also showed that processing history can be as important as chemical composition. The dramatic differences between furnace-cooled pearlitic steel and quenched martensitic steel illustrate how heat treatment can transform the mechanical behaviour of a material without altering its elemental composition. Consequently, successful material selection requires consideration of both composition and microstructure to ensure that engineering components perform safely and effectively throughout their service life.

11. CONCLUSION.

My investigation examined the microstructures of both ferrous and non-ferrous metallic materials and demonstrated how composition, processing history, grain boundaries and impurities influence mechanical properties. Through metallographic analysis and comparison with accredited reference sources, each specimen was successfully identified and its processing history determined from characteristic microstructural features.

One of the most significant findings of my investigation was the influence of processing history on material behaviour. Materials with similar chemical compositions exhibited substantially different properties as a result of different manufacturing and heat treatment processes. This was most clearly demonstrated by the comparison between Specimens D1 and D2. Although both specimens contained approximately 0.9% carbon, slow furnace cooling produced a pearlitic microstructure while rapid water quenching produced martensite. The martensitic structure exhibited significantly greater hardness and strength, illustrating how cooling rate can dramatically alter mechanical performance without changing composition.

My investigation also highlighted the importance of grain boundaries in controlling material strength and toughness. Grain boundaries act as barriers to dislocation movement and therefore contribute to strengthening mechanisms within metallic materials. Fine-grained structures contain a greater number of grain boundaries, increasing resistance to plastic deformation while also improving toughness by impeding crack propagation. The Hall-Petch relationship demonstrated the direct connection between grain size and yield strength, confirming the importance of grain refinement in engineering materials.

Impurities and inclusions were shown to have a considerable influence on material performance and failure behaviour. Sulphides, oxides and silicates can act as stress concentrators and potential crack initiation sites, particularly under cyclic loading conditions. The addition of manganese was shown to reduce the harmful effects of sulphur through the formation of manganese sulphides, improving the quality and reliability of structural steels. The investigation further demonstrated how impurity segregation at grain boundaries can contribute to embrittlement and increase susceptibility to brittle fracture.

A clear relationship was established between microstructure and mechanical properties throughout the investigation. Ferritic structures were associated with high ductility and formability, while increasing pearlite content produced greater strength and hardness. Martensitic structures provided maximum hardness and wear resistance but exhibited reduced ductility and toughness. The comparison between annealed and cold-worked brass showed how deformation processing increases strength through strain hardening, whereas annealing restores ductility through recrystallisation. Similarly, the comparison between ductile and grey cast iron demonstrated the importance of graphite morphology in controlling toughness and fracture resistance.

Based on the microstructural evidence and mechanical property evaluation, **Specimen C (Shipbuilding Steel)** offered the best overall balance of strength, toughness and ductility. Its ferrite-pearlite microstructure, combined with manganese strengthening and a hot-rolled processing history, provided a combination of mechanical properties suitable for demanding engineering applications. Unlike martensitic steel, it retained good toughness and fatigue resistance, while offering significantly greater strength than annealed low-carbon steel or brass. For this reason, Specimen C was judged to be the most versatile and effective engineering material examined during this investigation.

Overall, my investigation demonstrated that successful material selection depends not only on chemical composition but also on microstructure and processing history. Understanding these relationships allows engineers to predict material behaviour, optimise manufacturing processes and ensure the safe and reliable performance of engineering components in service.

12. REFERENCES.

I used the following sources to support material identification, metallurgical theory, phase analysis and engineering property evaluation.

Books.

Bolton, W. and Higgins, R.A. (Latest Edition). *Materials for Engineers and Technicians*. Routledge.

Academic and Technical Sources.

[DoITPoMS \(Dissemination of IT for the Promotion of Materials Science\)](#)

[MatWeb Material Property Data](#)

[MakeltFrom Material Database](#)

Course Documentation.

Pearson BTEC Higher Nationals in Engineering – Assignment Brief and Assessment Guidance.

Recommended Additional Reference.

[ASM International Materials Education Resources](#)

This source provides additional metallurgical information, phase diagrams, heat treatment data and engineering material references that support the findings of this investigation.